$u = k^T x + l r$

$\hat{x}$

$\dot{x} = Ax + bu$

$y = c^T x$

$\dot{\hat{x}} = A\hat{x} + bu + L(y - \hat{y})$

$\hat{y} = c^T \hat{x}$

$\dot{x} - \dot{\hat{x}} = \dot{\tilde{x}}$

$\dot{\tilde{x}} = Ax + \cancel{bu} - A\hat{x} - \cancel{bu} - L(y - \hat{y})$

$= A\tilde{x} - L(c^T x - c^T \hat{x})$

$- L(c^T \tilde{x})$

$\dot{\tilde{x}} = (A - Lc^T)\tilde{x}$

$u = k^T \hat{x} + l r = \Theta^T \omega$

$\dot{\hat{x}} = A\hat{x} + bu + (Ly + Lc^T\hat{x})$

$(A + Lc^T)\hat{x} + bu + Ly$

$\boxed{\hat{x} = ?}$

$\dot{\hat{x}} = \dots$

$s\hat{x} = (A - Lc^T)\hat{x} + bu + Ly$

$sI\hat{x} = (A - Lc^T)\hat{x} + bu + Ly$

$(sI - (A - Lc^T))\hat{x} = bu + Ly$

$\hat{x} = (sI - (A - Lc^T))^{-1}[bu + Ly]$

$\hat{x} = (sI - A + Lc^T)^{-1} bu + (\quad)^{-1} Ly$

$M^{-1} = \frac{1}{\det M}(\quad)$

$\begin{bmatrix} \frac{\bullet}{\Lambda(s)} & \frac{0}{\Lambda(s)} \\ \frac{\quad}{\Lambda(s)} & \frac{\times}{\Lambda(s)} \end{bmatrix}$

$\hat{x} = \left[\frac{1}{\Lambda(s)}\begin{bmatrix} \cdot & \cdot \\ \cdot & \cdot \end{bmatrix}\right] bu + \dots$

$\frac{\quad}{\Lambda(s)}$

$c^T \begin{bmatrix} s^{TREMA} \\ \vdots \\ s^0 \end{bmatrix}$

$\begin{bmatrix} s^{n-1} \\ \vdots \\ s^0 \end{bmatrix}$

MRAC
- grad ...
- stav ...

SYS
ZR $u = \Theta^T \omega$
RM

URO
y
ZA
$\dot{\Theta} = ?$

cieľ ?
URO = RM !!!
$\Theta^* ?$
P.Z.

$\dot{\Theta} = ?$
- grad (MIT)
- Lyap

zatiaľ $x = \sqrt{}$
$A \leftarrow$ frobo

$k > 0$

$\frac{y}{u} = \frac{k}{s}$ $\quad \frac{y_m}{r} = \frac{b_m}{s + a_m}$ $\quad b_m = a_m$

$u = \Theta(r - y)$

$\dot{\Theta} = ?$ grad! $\quad \dot{\Theta} = -\alpha\left(e \frac{\partial e}{\partial \Theta}\right)$ cit. fcn. ?

dá sa vôbec?

$e = y - y_m$
$e \to 0$

Θ

$\frac{\partial e}{\partial \Theta} = ?$

$e(\Theta) = "URO - RM"$
Θ

1.) !!!
P.Z.

URO ? $\quad \frac{y}{r} = ?$

$y = \frac{k}{s}(\Theta(r - y))$

$y = \frac{k\Theta}{s} r - \frac{k\Theta}{s} y$

$\frac{y_m}{r} = \frac{b_m}{s + a_m}$

$\frac{s + k\Theta}{s} y = \frac{k\Theta}{s} r$

$\frac{y}{r} = \frac{k\Theta}{s + k\Theta}$

P.Z. $\quad k\Theta^* = a_m$

$e = \overbrace{k\Theta \cdot (s + k\Theta)^{-1} r}^{y =} - y_m$

$\frac{\partial e}{\partial \Theta} = k(s + k\Theta)^{-1} r + k\Theta(-1)(s + k\Theta)^{-2}(k) r$

$= k(s + k\Theta) r - k^{-1}(s + k\Theta) \quad k\Theta^{-1}(s + k\Theta) r$

$\frac{\partial e}{\partial \Theta} = \frac{\quad}{A_m(s)}(r - y)$

$= k(s + k\Theta)^{-1}\left(r - \frac{k\Theta}{s + k\Theta} r\right)$

$$\hat{x} = \begin{bmatrix} \ddots \end{bmatrix} \frac{\alpha(s)}{\Lambda(s)} u + \dots$$

$$u = k^T \hat{x} + l r$$

$$u = k^T \left(\begin{bmatrix} \text{diag}_1 \end{bmatrix} \frac{\alpha(s)}{\Lambda(s)} u + \begin{bmatrix} \text{diag}_2 \end{bmatrix} \frac{\alpha(s)}{\Lambda(s)} y \right) + l r$$

$$u = \Theta_1^T \underbrace{\left(\frac{\alpha(s)}{\Lambda(s)} u \right)}_{\boxed{\nu_1}} + \Theta_2^T \underbrace{\left(\frac{\alpha(s)}{\Lambda(s)} y \right)}_{\nu_2} + \Theta_3 r$$

$$\Theta_1^T \nu_1 + \Theta_2^T \nu_2 + \Theta_3 r$$

$$u = \Theta^T \omega$$

$$y = k(s+k\Theta)^{-1} \left(r - \frac{k\Theta}{s+k\Theta} r \right)$$

$$\frac{\partial e}{\partial \Theta} = \frac{k}{s+k\Theta} \left[r - y \right]$$

aprox!!!

$$\frac{\partial e}{\partial \Theta} \approx \frac{\text{sign}(k)}{s + k\Theta^*} (r - y)$$

$$s + a_m$$

$$\dot{\Theta} = -\alpha e \left[\frac{\text{sign}(k)}{s + a_m} (r - y) \right]$$

MRAC stav

$e = x - x_m$

$\dot{x} = Ax + bu$

$\dot{e} = ?$ ← dyn. stav. adapt. odch.

$\dot{\Theta} = ?$

$\dot{x}_m = A_m x_m + b_m r$

$u = k^T x + l r$

$u = \Theta^T \omega$

1.) param SYS ← Θ^*

$\omega = \begin{bmatrix} x \\ r \end{bmatrix}$

$\Theta^T = [k^T \ l]$

!!!

URO

$$\dot{x} = Ax + b(k^T x + l r)$$

$$\dot{x} = (A + bk^T)x + bl r$$

$$\dot{x} = A_m x_m + b_m r$$

$$\dot{x} = Ax + b u$$

$$= Ax + bu + b(\overbrace{k^{*T} x + l^* r}^{u^*}) - b(k^{*T} x + l^* r)$$

$$= Ax + bk^{*T} x + bl^* r + b(u - \dots)$$

$$= \underbrace{(A + bk^{*T})}_{A_m} x + \underbrace{bl^*}_{b_m} r + b(\quad)$$

$$\boxed{\begin{aligned} A + bk^{*T} &= A_m \\ bl^* &= b_m \end{aligned}}$$

Θ^* ✓

$$\dot{x} = A_m x + b_m r + b(u - u^*)$$

$$\Theta^T \omega - \Theta^{*T} \omega$$

$$\theta^T \omega$$

$$\theta = \Theta - \Theta^*$$

1.)

$$\dot{x} = A_m x + b_m r + b\theta^T \omega \longleftarrow ?$$

2.) $\dot{e} = ?$

2) $\dot{e} = ?$

$e = x - x_m$

$\dot{e} = \dot{x} - \dot{x}_m$

$\dot{e} = A_m x + \cancel{b_m r} + b\theta^T\omega - A_m x_m - \cancel{b_m r}$

$\dot{e} = A_m (x - x_m) + b(\theta^T\omega)$

$\dot{e} = A_m e + b(\theta^T\omega) \longleftarrow ?$

✓

MRC uni riešiť ?

x ✓

$u = k^T x + l r$ ✓

??? nemám!

real odhad

$x \leftarrow \hat{x}$

$x(0) = x_0$

$\dot{x} = Ax + bu$

$y = c^T x$

$\dot{\hat{x}} = A\hat{x} + bu + L(y - \hat{y})$

$\hat{y} = c^T \hat{x}$

$\hat{x}(0) = \hat{x}_0$

$y = \hat{y}$

$y - \hat{y} = 0$

$\tilde{x} = x - \hat{x} \longrightarrow 0$

$\dot{\tilde{x}} = \dot{x} - \dot{\hat{x}}$

$\tilde{x}(0) = \tilde{x}_0$

$\dot{\tilde{x}} = Ax + \cancel{bu} - A\hat{x} - \cancel{bu} - L(y - \hat{y})$

$\dot{\tilde{x}} = A(x - \hat{x}) - L(c^T x - c^T \hat{x})$

$\dot{\tilde{x}} = A\tilde{x} \ldots$

$$x = A(x - \hat{x}) - L(c^T x - c^T \hat{x})$$

$$\dot{\tilde{x}} = A\tilde{x} - Lc^T(\tilde{x})$$

$$\boxed{\dot{\tilde{x}} = (A - Lc^T)\tilde{x}}$$

$$\tilde{x} \to 0 \qquad \tilde{x}(0) = \tilde{x}_0$$

$$\boxed{\dot{x} = Ax}$$